

# Check of the Updated Shortcut Calculation

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## Summary

I checked the updated TeX file against the direct finite stochastic shortcut matrix. The leading corrected resolvent formula now has the stochastic shortcut sign, but the correction has not been propagated consistently into the first-passage formula. In particular, the equations labelled `first_pass_prob` and `first_pass_prob.2` still need the corresponding sign update to match the corrected resolvent and the finite-matrix benchmark.

## First-passage sign check

For a shortcut that moves probability mass

$$\lambda = \beta(1 - q)$$

from the self-loop  $u \rightarrow u$  to the absorbing target edge  $u \rightarrow v$ , the transient dynamics loses this mass at  $u$ . Summing the corrected absorbing propagator gives

$$\tilde{F}_{n_0}(v, z) = \tilde{\mathcal{F}}_{n_0}(v, z) + \frac{z\beta(1 - q)\tilde{W}_{n_0}(u, z) \left[1 - \tilde{\mathcal{F}}_u(v, z)\right]}{1 + z\beta(1 - q)\tilde{W}_u(u, z)}.$$

Equivalently, with

$$\tilde{H}_{N,|u-v|}(z) = \frac{\beta(1 - q)}{q} \frac{1}{1 + z\beta(1 - q)\tilde{W}_u(u, z)},$$

the prefactor in the next displayed first-passage equation should be positive:

$$\tilde{F}_{n_0}(v, z) = \tilde{\mathcal{F}}_{n_0}(v, z) + qz\tilde{W}_{n_0}(u, z) \left[1 - \tilde{\mathcal{F}}_u(v, z)\right] \tilde{H}_{N,|u-v|}(z).$$

As currently displayed, the first expression has a minus denominator and the second expression has a minus prefactor. These two forms therefore do not yet match the corrected  $H$  definition.

## Finite-matrix benchmark

I used the direct stochastic matrix with  $N = 100$ ,  $q = 2/3$ ,  $\beta = 0.04$ , shortcut  $u = 6 \rightarrow v = 56$  in paper indexing, and target  $v = 56$ . The check used  $n_0 = 1, \dots, 5$  and  $z \in \{0.1, 0.3, 0.6, 0.85, 0.95\}$ .

| Formula compared with the direct stochastic matrix     | max absolute difference |
|--------------------------------------------------------|-------------------------|
| correct plus/plus first-passage formula                | $2.08 \times 10^{-17}$  |
| updated <code>first_pass_prob</code> denominator sign  | $3.29 \times 10^{-3}$   |
| updated <code>first_pass_prob.2</code> minus prefactor | $6.34 \times 10^{-2}$   |

## Pole comparison

For the symmetric case  $|u - v| = N/2$ , the updated expression has

$$\tilde{H}_{N,N/2}(z) = \frac{1}{\frac{q}{\beta(1-q)} + \phi(z)}.$$

Writing  $s = 1/z$ , for  $N = 100$ ,  $q = 2/3$ ,

$$\gamma_1 = 1 - q + q \cos \frac{2\pi}{N} = 0.9986844856,$$

and

$$\alpha_1 = 1 - q + q \cos \frac{\pi}{N} = 0.9996710402.$$

The largest pole  $s_1$  solves

$$\frac{q}{\beta(1-q)} + \phi(1/s) = 0.$$

At  $s = \gamma_1$ , the  $\phi(1/s)$  term is zero, so the denominator is  $q/[\beta(1-q)] > 0$  for every finite positive  $\beta$ . Thus  $s_1$  cannot equal  $\gamma_1$  for  $0 < \beta \leq 1$ . Numerically,

| $\beta$ | $s_1$        | $s_1 - \gamma_1$      | $\alpha_1 - s_1$      |
|---------|--------------|-----------------------|-----------------------|
| 0.01    | 0.9996076314 | $9.23 \times 10^{-4}$ | $6.34 \times 10^{-5}$ |
| 0.04    | 0.9994512449 | $7.67 \times 10^{-4}$ | $2.20 \times 10^{-4}$ |
| 0.08    | 0.9993014753 | $6.17 \times 10^{-4}$ | $3.70 \times 10^{-4}$ |
| 1.00    | 0.9987831951 | $9.87 \times 10^{-5}$ | $8.88 \times 10^{-4}$ |

So the proposed switch at  $s_1 = \gamma_1$  does not occur in the admissible range  $0 < \beta \leq 1$  for this corrected  $H$ .

## Double-peak scan

Using the exact first-passage PMF for the same  $N = 100$ ,  $q = 2/3$ ,  $u = 6 \rightarrow v = 56$ , target  $v = 56$  setup, the result depends on what is meant by “double peak.” With the report’s clear visual rule

$$h_v / \min(h_1, h_2) \leq 0.8,$$

with two filtered peaks and peak separation at least 10, the scan gives:

| $n_0$ | clear double peak? | largest checked $\beta$ with clear double peak |
|-------|--------------------|------------------------------------------------|
| 1     | yes                | 0.030                                          |
| 2     | yes                | 0.030                                          |
| 3     | yes                | 0.030                                          |
| 4     | no                 | —                                              |
| 5     | no                 | —                                              |
| 6     | no                 | —                                              |

At  $\beta_c = 2q/[(1-q)N] = 0.04$ , no case remains clearly double-peaked under this rule; for  $n_0 = 1, 2, 3$  the inter-peak valley is already shallow,

$$h_v / \min(h_1, h_2) \approx 0.887\text{--}0.891.$$

If one only asks for two local maxima, the boundary is less strict and local maxima persist to about  $\beta = 0.060\text{--}0.062$  for  $n_0 = 1, 2, 3$ , but the shape is no longer a clear two-peak distribution.